

Pathway Enrichment on the Four Retained Gene Programs (K=7)

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1. What changed since the July report, and why

The July 15 report characterized the two **survival-active** gene programs of the recommended D4 fit (Program 3, protective; Program 7, adverse). Its conclusions on those two programs are unchanged here, and for a specific reason: **the underlying fit is the same fit**. The K -selection procedure was reworked in the 8/19–8/20 cycle from cross-validated external C-index to ARD pruning plus ELBO comparison, but that procedure *re-confirmed* $K = 7$ on the same D4 configuration rather than producing a different model (DECISIONS.md 2026-08-19). Programs 3 and 7 are still the only two programs carrying a non-negligible survival coefficient.

What did change is the **classification of the other five programs**. The July analysis treated them as a single undifferentiated “Inactive” bucket. The three-way classification introduced with `classify_factors()` splits that bucket:

Class	Criterion	Programs
survival-active	$ \tilde{\beta}_k > 0.001$	3, 7
genomics-only	not survival-active, but $\text{PVE}_k > 1\%$	5, 6
dead	neither; fully pruned by ARD	1, 2, 4

Programs 5 and 6 are therefore real gene-expression programs, not noise, and the retained set is four factors rather than two. They were never characterized. In fact their enrichment had already been *computed* under the old convention and then discarded at the reporting step, because every output table was filtered to the two survival-active programs. Program 6 had 40 significant gene sets and Program 5 had 12, all silently dropped.

This report extends every analysis in the July report to the four retained programs. The practical question it answers: **are the two genomics-only programs interpretable biology, and does their presence change how the model should be described?**

Direction convention (unchanged)

Adverse/protective labels come from the *marginal* survival association of each program’s projection score (Cox on ZF_k alone), not from the joint model’s $\tilde{\beta}$ sign. For Programs 3 and 7 the two disagree ($\hat{\beta}_7 = -0.041$, $\hat{\beta}_3 = +0.011$), a suppression effect between correlated programs in the joint fit. The marginal convention is used throughout (DECISIONS.md 2026-06-16). The genomics-only programs have $\tilde{\beta}_k \approx 0$ by definition and carry no direction label.

2. Methods

Unchanged from the July report except for scope. Gene weights are the posterior means $\bar{f}_{jk}$ from the D4 fit (2,064 genes $\times$ 7 programs); the `point_exponential` prior makes them non-negative, so $|\bar{f}_{jk}| = \bar{f}_{jk}$ and programs are ranked by raw weight.

- **fgsea** (primary): genes ranked by weight within a program, run against Hallmark, Reactome, KEGG, GO-BP, and a custom PDAC collection (Moffitt/Bailey subtype signatures plus DeSurv’s three published factor gene lists). Significance at $\text{padj} < 0.10$ (Benjamini–Hochberg within program $\times$ collection), `seed = 1`.
- **ORA** (sensitivity): hypergeometric over-representation on the top N genes, $N \in \{50, 100, 150\}$, background = the 2,064-gene selected universe.
- **Subtype concordance**: Spearman correlation and Kruskal–Wallis test of each program’s patient loadings against the PurlST classical/basal score in TCGA-PAAD ($n = 144$).
- **DeSurv overlap**: Jaccard index and hypergeometric p against each of DeSurv’s three published factors, top 270 genes per program, restricted to the shared background.
- **External cohort robustness**: each program’s headline leading-edge signature scored in each of the 5 held-out cohorts, then a univariate Cox fit per cohort. For Programs 3 and 7 this is confirmatory. **For Programs 5 and 6 it is a negative control** (see §5).

3. Enrichment across the four retained programs

Table 2: Significant gene sets per program, all 7 programs. The two programs previously labeled 'Inactive' with the most enrichment (5 and 6) are exactly the two now classified genomics-only; the three dead programs return zero sets, as expected.

Program	Label	Class	Sig. sets (padj<0.10)
1	Dead	dead	0
2	Dead	dead	0
3	Protective	survival_active	6
4	Dead	dead	0
5	Genomics-only	genomics_only	12
6	Genomics-only	genomics_only	40
7	Adverse	survival_active	7

The three dead programs return **zero** significant sets, while both genomics-only programs return many. That is a useful internal consistency check on the classification: the PVE-based split between “genomics-only” and “dead” was made without reference to gene sets, yet it lands exactly on the programs that have coherent pathway signal.

Table 3: Top 6 gene sets per retained program by adjusted p-value (fgsea). Full table: T5_enrichment_kept.csv.

Program	Class	Collection	Set	NES	padj
Program 3	survival_active	PDAC_custom	DeSurv D1 ClassicalTumor	1.06	1.7e-02
Program 3	survival_active	PDAC_custom	DeSurv D3 BasalLikeTumor	1.06	1.7e-02
Program 3	survival_active	KEGG	ITGA B FAK CDC42 SIGNALING PATHWAY	1.28	2.3e-02
Program 3	survival_active	KEGG	ITGA B FAK RAC SIGNALING PATHWAY	1.28	2.3e-02
Program 3	survival_active	KEGG	ITGA B RHOGAP RHOA SIGNALING PATHWAY	1.28	2.3e-02
Program 3	survival_active	PDAC_custom	Moffitt Classical	1.17	3.5e-02
Program 5	genomics_only	Hallmark	EPITHELIAL MESENCHYMAL TRANSITION	1.09	5.5e-04
Program 5	genomics_only	PDAC_custom	DeSurv D2 StromalImmune	1.06	9.1e-04
Program 5	genomics_only	Reactome	EXTRACELLULAR MATRIX ORGANIZATION	1.08	1.0e-02
Program 5	genomics_only	GO_BP	CELL ADHESION	1.05	2.2e-02
Program 5	genomics_only	GO_BP	TUBE DEVELOPMENT	1.06	2.2e-02
Program 5	genomics_only	GO_BP	ANATOMICAL STRUCTURE FORMATION INVOLVED IN MORPHOGENESIS	1.05	5.0e-02
Program 6	genomics_only	PDAC_custom	DeSurv D1 ClassicalTumor	1.06	8.0e-03
Program 6	genomics_only	PDAC_custom	DeSurv D2 StromalImmune	1.07	8.0e-03
Program 6	genomics_only	GO_BP	CELL ACTIVATION	1.08	2.7e-02
Program 6	genomics_only	GO_BP	LYMPHOCYTE ACTIVATION	1.11	2.7e-02
Program 6	genomics_only	GO_BP	POSITIVE REGULATION OF CELL CELL ADHESION	1.15	2.7e-02
Program 6	genomics_only	GO_BP	REGULATION OF CELL POPULATION PROLIFERATION	1.07	2.7e-02
Program 7	survival_active	PDAC_custom	DeSurv D3 BasalLikeTumor	1.08	6.1e-04
Program 7	survival_active	KEGG	GF RTK PI3K SIGNALING PATHWAY	1.23	1.1e-03
Program 7	survival_active	KEGG	GF RTK RAS PI3K SIGNALING PATHWAY	1.23	1.1e-03
Program 7	survival_active	KEGG	GF RTK RAS ERK SIGNALING PATHWAY	1.23	1.2e-03
Program 7	survival_active	Reactome	DEVELOPMENTAL CELL LINEAGES	1.13	1.0e-02
Program 7	survival_active	PDAC_custom	Bailey Squamous	1.16	3.0e-02

What the two genomics-only programs look like

The top-weighted genes are unusually legible for both:

Table 4: Top 20 genes by posterior mean weight for the two genomics-only programs.

Program	Top 20 genes by weight
Program 5	COL1A2, COL5A2, COL3A1, COL12A1, THBS2, HTRA1, CDH11, SPARC, ANTXR1, FZD1, PXDN, TIMP2, AEBP1, CTHRC1, COL5A1, COL6A3, C1QTNF3, COL8A1, NUA1, FAP
Program 6	CELF2, IKZF1, RCSD1, AKNA, FZD4, IL10RA, PREX1, COL14A1, RASSF2, SMAP2, APOLD1, TNXB, FBLN5, ITPR1, DOCK10, SLIT3, WDFY4, CHRDL1, HCLS1, SPARCL1

Program 5 reads as desmoplastic stroma / cancer-associated fibroblast. Seven collagen genes appear in the top 20 (COL1A2, COL3A1, COL5A1, COL5A2, COL6A3, COL8A1, COL12A1), alongside SPARC, THBS2, TIMP2, CTHRC1, CDH11, AEBP1, and **FAP**, the canonical fibroblast-activation marker. Its enrichment agrees: Hallmark epithelial-mesenchymal transition is the top set, followed by DeSurv's StromalImmune factor and Reactome extracellular matrix organization, then cell adhesion and tube/vessel development.

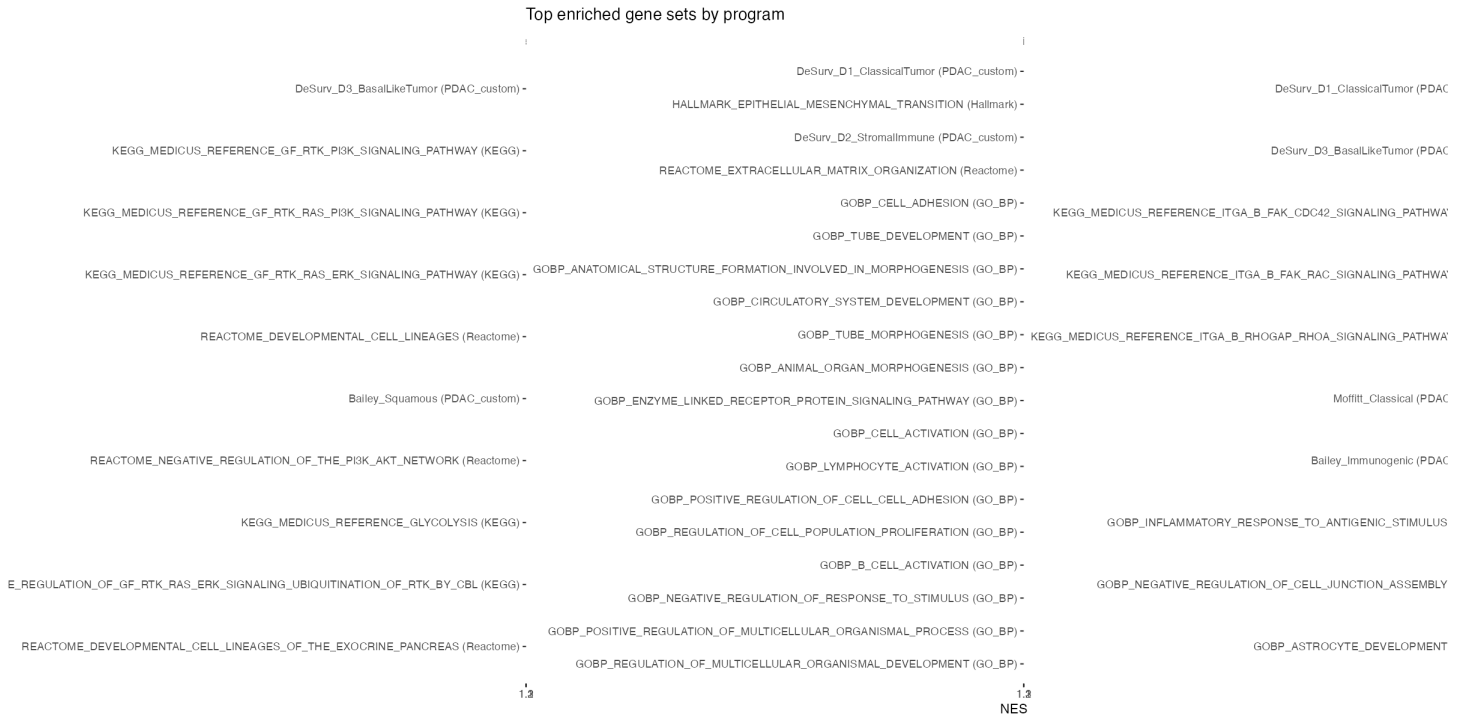


Figure 1: Enrichment dotplot across the four retained programs (F1). Each program's top sets by adjusted p-value; dot size is set size, color is normalized enrichment score.

Program 6 reads as immune infiltrate, with a vascular component. The top genes are predominantly lymphoid (IKZF1, AKNA, IL10RA, HCLS1, DOCK10, WDFY4, CELF2, RCSD1, PREX1) mixed with vascular and basement-membrane genes (FBLN5, SLIT3, SPARCL1, APOLD1, COL14A1, TNXB, CHRDL1). Its enrichment is correspondingly immune-dominated: cell activation, lymphocyte activation, B-cell activation, and positive regulation of cell-cell adhesion.

Neither reading is a formal claim. Both are what the gene lists and enrichment tables say on their face, and both are consistent across two independent methods (fgsea ranking and the DeSurv overlap in §6). See §7 for the standing caveat about methodology.

4. Subtype concordance

Table 5: Concordance of program loadings with the PurlST basal/classical axis, TCGA-PAAD (n=144). Positive rho = basal-leaning.

Program	Label	Spearman rho	Spearman p	Kruskal p
Program 3	Protective	-0.575	4.9e-14	5.1e-07
Program 5	Genomics-only	+0.299	2.8e-04	5.6e-01
Program 6	Genomics-only	-0.026	7.5e-01	6.8e-03
Program 7	Adverse	+0.582	2.1e-14	3.2e-12

The two survival-active programs sit at opposite ends of the subtype axis, strongly and symmetrically (Program 3 $\rho = -0.575$, Program 7 $\rho = +0.582$). This is the subtype-survival axis the PDAC literature already describes, and it is the July report's central result.

The genomics-only programs behave differently, and consistently with a stromal/immune reading. **Program 6 is essentially orthogonal to the subtype axis** ($\rho = -0.026$, $p = 0.75$), which is what an immune-infiltrate program should look like: tumor-cell subtype and immune content are separate axes of variation. **Program 5 shows a weak basal lean** ($\rho = +0.299$, $p = 2.8 \times 10^{-4}$), which is also expected, since basal-like PDAC tumors are the more stroma-rich ones. Note Program 6's significant Kruskal-Wallis test alongside its null Spearman: the relationship with subtype *groups* is non-monotonic, so it is not captured by a rank correlation.

5. External cohort robustness (negative control for the genomics-only pair)

Each program's headline leading-edge signature is scored in each held-out cohort, then fit in a univariate Cox model. For Programs 3 and 7, replicating the hazard-ratio direction across independent cohorts is evidence their prognostic signal is real rather than a training-set artifact. For Programs 5 and 6 the logic runs the other way: they carry no survival coefficient in the joint fit, so a null result *supports* the genomics-only label, and a consistently significant one would mean the label understates them.

Table 6: Univariate Cox HR for each program's headline (lowest-padj) signature, per held-out cohort. Note the Signature column for Program 6.

Program	Label	Signature	Cohort	n	HR	p
Program 3	Protective	DeSurv D1 ClassicalTumor	Dijk	90	0.84	3.3e-01
Program 3	Protective	DeSurv D1 ClassicalTumor	Moffitt_GEO_array	123	0.88	5.5e-01
Program 3	Protective	DeSurv D1 ClassicalTumor	PACA_AU_array	63	0.49	3.6e-02
Program 3	Protective	DeSurv D1 ClassicalTumor	PACA_AU_seq	52	0.42	1.2e-02
Program 3	Protective	DeSurv D1 ClassicalTumor	Puleo_array	288	0.68	7.4e-03
Program 5	Genomics-only	EPITHELIAL MESENCHYMAL TRANSITION	Dijk	90	1.50	6.7e-03
Program 5	Genomics-only	EPITHELIAL MESENCHYMAL TRANSITION	Moffitt_GEO_array	123	0.98	9.1e-01
Program 5	Genomics-only	EPITHELIAL MESENCHYMAL TRANSITION	PACA_AU_array	63	1.20	4.4e-01
Program 5	Genomics-only	EPITHELIAL MESENCHYMAL TRANSITION	PACA_AU_seq	52	0.73	2.8e-01
Program 5	Genomics-only	EPITHELIAL MESENCHYMAL TRANSITION	Puleo_array	288	1.15	1.7e-01
Program 6	Genomics-only	DeSurv D1 ClassicalTumor	Dijk	90	1.29	1.6e-01
Program 6	Genomics-only	DeSurv D1 ClassicalTumor	Moffitt_GEO_array	123	0.90	5.9e-01
Program 6	Genomics-only	DeSurv D1 ClassicalTumor	PACA_AU_array	63	0.53	9.9e-02
Program 6	Genomics-only	DeSurv D1 ClassicalTumor	PACA_AU_seq	52	0.22	6.2e-04
Program 6	Genomics-only	DeSurv D1 ClassicalTumor	Puleo_array	288	0.68	1.7e-02
Program 7	Adverse	DeSurv D3 BasalLikeTumor	Dijk	90	1.99	4.6e-03
Program 7	Adverse	DeSurv D3 BasalLikeTumor	Moffitt_GEO_array	123	1.30	1.9e-01
Program 7	Adverse	DeSurv D3 BasalLikeTumor	PACA_AU_array	63	1.70	2.2e-01
Program 7	Adverse	DeSurv D3 BasalLikeTumor	PACA_AU_seq	52	2.40	7.3e-02
Program 7	Adverse	DeSurv D3 BasalLikeTumor	Puleo_array	288	1.56	4.8e-03

Programs 3 and 7 replicate. Program 3's signature is protective in all 5 cohorts (HR 0.42–0.88, 3 of 5 significant); Program 7's is adverse in all 5 (HR 1.30–2.40, 2 of 5 significant). Consistent direction across every cohort, with significance where sample size allows.

The negative control as first run is confounded for Program 6, and the confound is instructive. The runner defines a program's headline signature as its single lowest-padj enriched set. For Program 6 that set is DeSurv_D1_ClassicalTumor, the *same* gene list that serves as Program 3's signature. Scoring it measures the classical-tumor axis's prognostic value, not whether Program 6 carries survival signal, and it duly came back protective-leaning (HR < 1 in 4 of 5 cohorts, 2 significant). Read naively, that looks like evidence against the genomics-only label. It is really just the classical signature reappearing.

Re-running the control on each program's own pathway character removes the confound. Excluding the PDAC_custom collection (DeSurv's three published factor lists, the known subtype/prognostic axes) and taking each genomics-only program's top remaining set gives EMT for Program 5 and GOBP_CELL_ACTIVATION for Program 6:

- Program 5: 1/5 cohorts significant at $p < 0.05$; HR range 0.73--1.50; 3/5 with $HR > 1$.

- Program 6: 0/5 cohorts significant at $p < 0.05$; HR range 0.53--1.39; 2/5 with $HR > 1$.

Program 6's immune signature is null: **0 of 5 cohorts significant**, HR straddling 1 in both directions (0.53–1.39). The entire apparent protective effect in the first table was the borrowed classical-tumor gene list. Program 5's EMT signature is also essentially null, with 1 of 5 cohorts significant (Dijk, HR 1.50) and directions mixed across the rest. No consistent prognostic effect.

The negative control passes for both genomics-only programs, which is the label-confirming result. It also carries a methodological lesson worth keeping: "lowest padj" is a poor definition of a program's headline signature when the gene-set universe includes signatures that are themselves prognostic axes. Any future factor added to the retained set should be scored on its own distinctive pathway character, not on whichever set happens to rank first.

6. Overlap with DeSurv's published factors

Table 7: Gene-list overlap with DeSurv's three published factors (top 270 genes per program).

Program	Label	DeSurv factor	Jaccard	Hypergeometric p
Program 3	Protective	ClassicalTumor	0.133	1.1e-07
Program 3	Protective	BasalLikeTumor	0.102	2.8e-03

Program	Label	DeSurv factor	Jaccard	Hypergeometric p
Program 3	Protective	StromalImmune	0.006	1.0e+00
Program 5	Genomics-only	StromalImmune	0.179	< 1e-15
Program 5	Genomics-only	ClassicalTumor	0.028	1.0e+00
Program 5	Genomics-only	BasalLikeTumor	0.019	1.0e+00
Program 6	Genomics-only	StromalImmune	0.166	7.0e-14
Program 6	Genomics-only	ClassicalTumor	0.109	3.2e-04
Program 6	Genomics-only	BasalLikeTumor	0.002	1.0e+00
Program 7	Adverse	BasalLikeTumor	0.178	< 1e-15
Program 7	Adverse	StromalImmune	0.045	9.9e-01
Program 7	Adverse	ClassicalTumor	0.022	1.0e+00

This is the most informative single result in the report. Each of our four retained programs maps onto exactly one of DeSurv's three factors:

- Program 3 → **ClassicalTumor** ($J = 0.133$, $p \approx 1 \times 10^{-7}$)
- Program 7 → **BasalLikeTumor** ($J = 0.178$, $p \approx 3 \times 10^{-16}$)
- Program 5 → **StromalImmune**, and only StromalImmune ($J = 0.179$, $p \approx 9 \times 10^{-17}$; the other two DeSurv factors give $p = 1.0$)
- Program 6 → **StromalImmune** ($J = 0.166$, $p \approx 7 \times 10^{-14}$), with a secondary ClassicalTumor overlap ($J = 0.109$, $p \approx 3 \times 10^{-4}$)

So the four-factor retained set recovers DeSurv's full three-factor structure, with DeSurv's single combined StromalImmune factor split across two of our programs. Given the gene content in §3, that split is along stroma (Program 5, collagen/CAF) versus immune (Program 6, lymphoid) lines.

This reframes what the model is doing. Under the old two-program description, the model found the tumor-intrinsic prognostic axis and discarded everything else. In fact it also recovers the tumor microenvironment as structure, resolved at finer grain than the published comparison, and correctly assigns it no prognostic weight of its own once the basal/classical axis is accounted for. That is a more complete and more defensible account of the fit, and it also explains *why* the genomics-only programs exist: stromal and immune content drive a large share of bulk expression variance in PDAC, so a factorization must represent them, but they are not the prognostic axis in this model.

7. Conclusions and standing caveat

1. The July report's findings on Programs 3 and 7 stand unchanged. The same fit, the same two survival-active programs, the same adverse/basal and protective/classical identification, now supported by the same five methods.
2. The two genomics-only programs are interpretable, not residual noise. Program 5 reads as desmoplastic stroma / CAF, Program 6 as immune infiltrate with a vascular component. Their enrichment, gene content, subtype behavior, and DeSurv overlap all agree.
3. The retained set recovers DeSurv's full three-factor structure, splitting DeSurv's combined StromalImmune factor into separate stromal and immune programs.
4. The three dead programs return zero enriched sets, independently corroborating the PVE-based genomics-only/dead split.
5. Both genomics-only programs pass the external-cohort negative control once scored on their own pathway character rather than on a borrowed tumor-subtype signature (§5). Getting this right required correcting a confound in how a program's "headline" signature is chosen; the uncorrected version made Program 6 look prognostic when it is not.

Caveat on methodology, not on evidence. This line of investigation was pursued on our own initiative rather than at the group's direction, and the specific methods here (fgsea/ORa against these collections, PurlST concordance, DeSurv overlap, per-cohort hazard ratios) were chosen for availability rather than from a deliberate review of what the question requires. The results above are internally consistent and mutually corroborating, and may well be sufficient as they stand. Confirming that the approach is the right one is worth doing before these biological labels are carried into the manuscript.

Open follow-up, unchanged from the 8/21 chapter: the cross-cohort *consistency* of the two genomics-only programs has not been tested the way Programs 3 and 7 have. §5 is a prognostic negative control, not a test of whether the same stromal and immune programs reappear when the model is refit on other cohorts.

